

Program : Diploma in Robotic Process Automation	
Course Code : 4308	Course Title: Web Scripting Language Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To implement effective webpages using HTML and DOM..
- To implement client side interface using JAVASCRIPT and JQuery .
- To familiarize the functionalities of Ajax and Angular JS .

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic programming skills in C language.	2339	Introduction to Computation and Programming	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Build Webpages using HTML.	10	Applying
CO2	Utilize JavaScript to add functionality to webpages.	10	Applying
CO3	Implement webpages using DOM.	12	Applying
CO4	Implement different Ajax, JQuery & Angular JS functionalities in Web development.	10	Applying
	Series Test	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	1		3	3			
CO2	2	3	3	3			
CO3	3	3	3	3			
CO4	3	3	3	3			

3 – Strongly mapped

2 – Moderately mapped

1 – Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Build Webpages using HTML.		
M1.01	Install, setup Integrated Development Environment (IDE) for web development.	1	Applying
M1.02	Create a web page with all possible elements of HTML.	2	Applying
M1.03	Create a Responsive Web page with HTML.	3	Applying
M1.04	Display various forms of XML documents.	3	Applying
CO2	Utilize JavaScript to add functionality to webpages.		
M2.01	Programs to demonstrate JavaScript array, object and functions.	3	Applying
M2.02	Client Side Scripts for Form Validation using JavaScript.	3	Applying
M2.03	Programs to demonstrate JavaScript decision making statements & loops.	4	Applying
	Series Test 1	1.5	
CO3	Implement webpages using DOM.		
M3.01	Programs to demonstrate DOM.	2	Applying
M3.02	Programs to perform DOM Traversing.	2	Applying
M3.03	Programs to demonstrate event handling	3	Applying

M3.04	Implementation of set operators, nested queries and Join queries.	3	Applying
M3.05	Creation of Views, Cursors and Triggers	2	Applying
CO4	Implement different Ajax, JQuery & Angular JS functionalities in Web development.		
M4.01	Programs to familiarize JQUERY	3	Applying
M4.02	Creating a responsive webpage using the functionalities of AJAX ,JQUERY& Angular JS	3	Applying
M4.03	Design website for a specific application.	4	Applying
	Series Test 2	1.5	

Text / Reference:

T/R	Book Title/Author
T1	Paul J. Deitel, Harvey M. Deitel, Abbey Deitel, “ Internet and World Wide Web How To Program ”, 5/E, Pearson Education, 2012.
R1	Jon Duckett , “ HTML and CSS: Design and Build Websites ”, Wiley
R2	Jon Duckett , “ JavaScript and JQuery : Interactive Front–End Web Development ”, Wiley.
R3	Felipe Coury,Ari Lerner, Carlos Taborda”ng – book: The Complete Guide to Angular Paperback - 6 ”, February 2018.

Online Resources:

Sl.No	Website Link
1	https://www.w3schools.com/html/default.asp
2	https://www.tutorialspoint.com/html5/index.htm
3	https://www.w3schools.com/js/default.asp
4	https://www.w3schools.com/angular

List of Sample Experiments.

1. Install, setup Integrated Development Environment (IDE) for web development.
2. Implement Basic HTML Tags
3. Implement Table Tags & Frames.
4. Design a Form in HTML.

5. Implement a web page using HTML.
6. Implement Basic XML Tags.
7. Display Raw XML Document.
8. Implement JavaScript programs to add numbers.
9. Implement JavaScript programs to swap two variables.
10. Implement JavaScript program to solve quadratic equation.
11. Implement JavaScript program to calculate the area of a triangle.
12. Implement JavaScript program to print all prime numbers in an interval.
13. Implement JavaScript program to check Armstrong number in an interval.
14. Implement JavaScript program to check if the numbers have same last digit.
15. Implement JavaScript program to shuffle deck of cards.
16. Implement JavaScript program to check a key exist in an object.
17. Implement JavaScript program to count the number of properties in an object.
18. Implement JavaScript program to replace all instances of character in a string.
19. Implement JavaScript programs to convert decimal to binary.
20. Programs to demonstrate DOM and event handling
21. Programs to familiarize JQuery & Ajax.
22. Application program to familiarize Angular JS